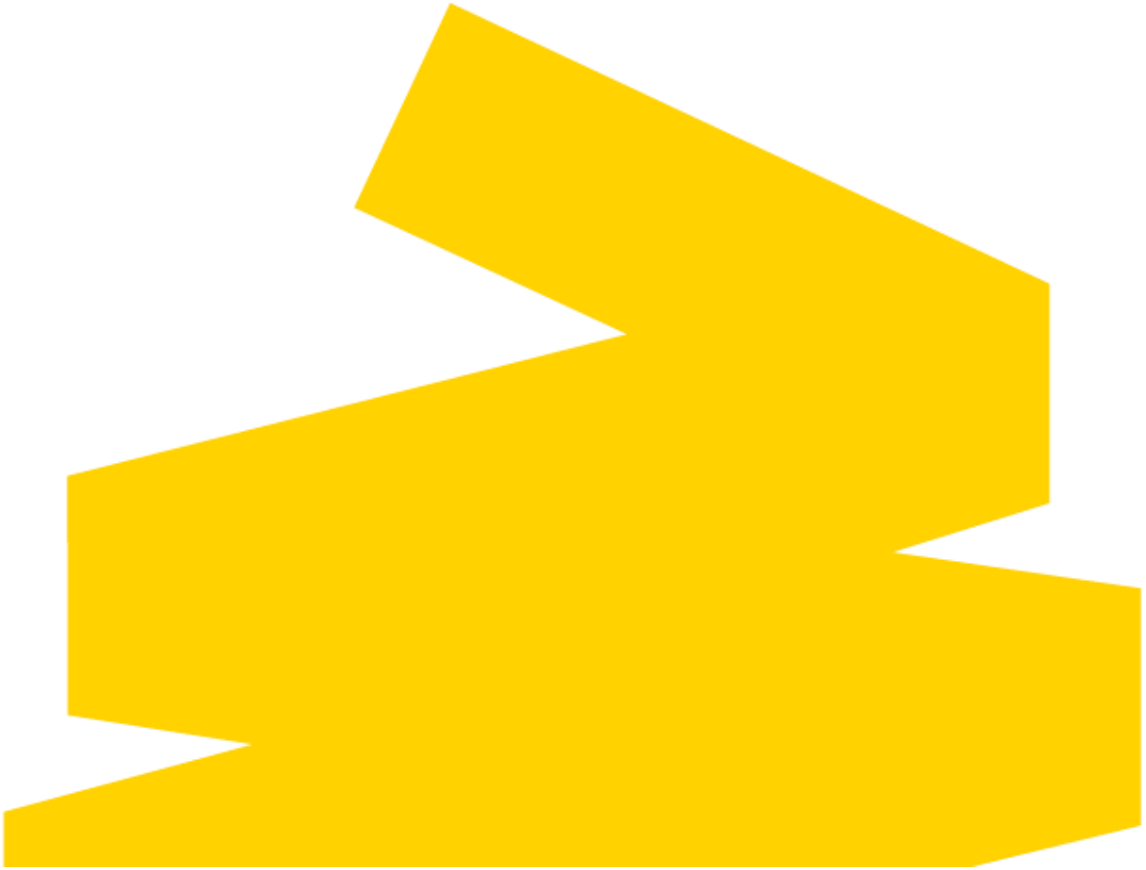
TURKU AMK

Windows server security enhancement
Janne Turpeinen

Windows server security enhancement



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1 Introduction

Purpose of this report is to show how the windows server 2022 security is improved by using security enhancements. Security enhancements that were applied are restricting user logons to specific named devices, logon hours for users and logon auditing.

2 Security enhancements

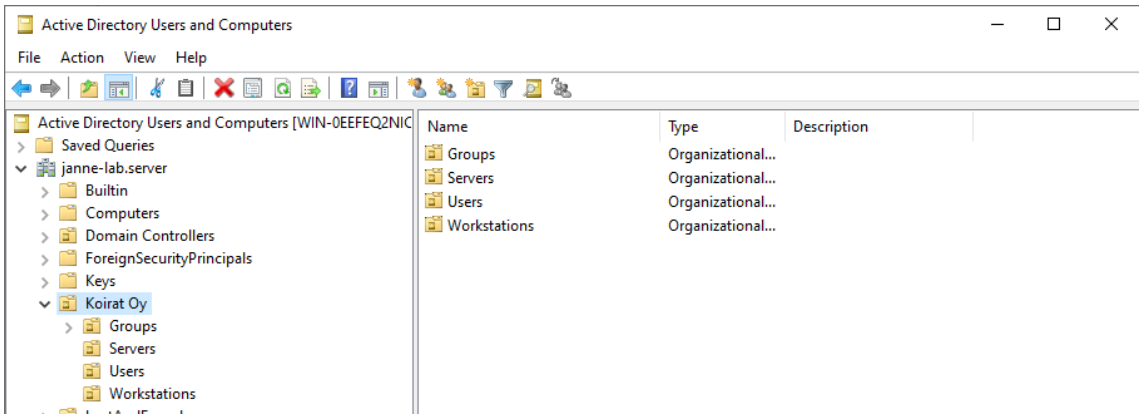
In this work the security enhancements what were applied are: **restricting user logons to specific named devices**, **logon hours for users** and **logon auditing**. Also using the **Group Policy** and **Organizational Units** are part of the security.

Group Policy means that in Active Directory, Windows settings are defined for each group so that they are applied automatically to workstations when a user logs in. This way, settings do not need to be configured individually for each computer.

Organizational Units are a way in Active Directory to manage users, groups and devices effortlessly.

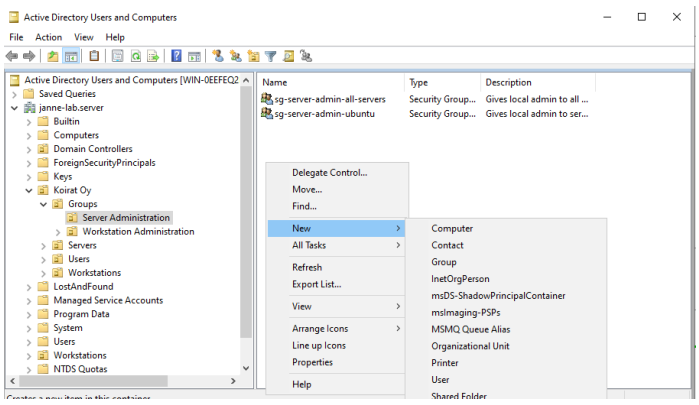
In the beginning you make new organizational unit for your company in the Active Directory. This is accessed by using Server Manager and pressing Tools from the right upper corner and there you choose Active Directory users and Computers. You can make new organizational units by right clicking or from the upper tool bar.

In the example the new organizational unit called Koirat Oy, were created, and inside this organizational unit, four new organizational units were made which is called Groups, Servers, Users and Workstations.



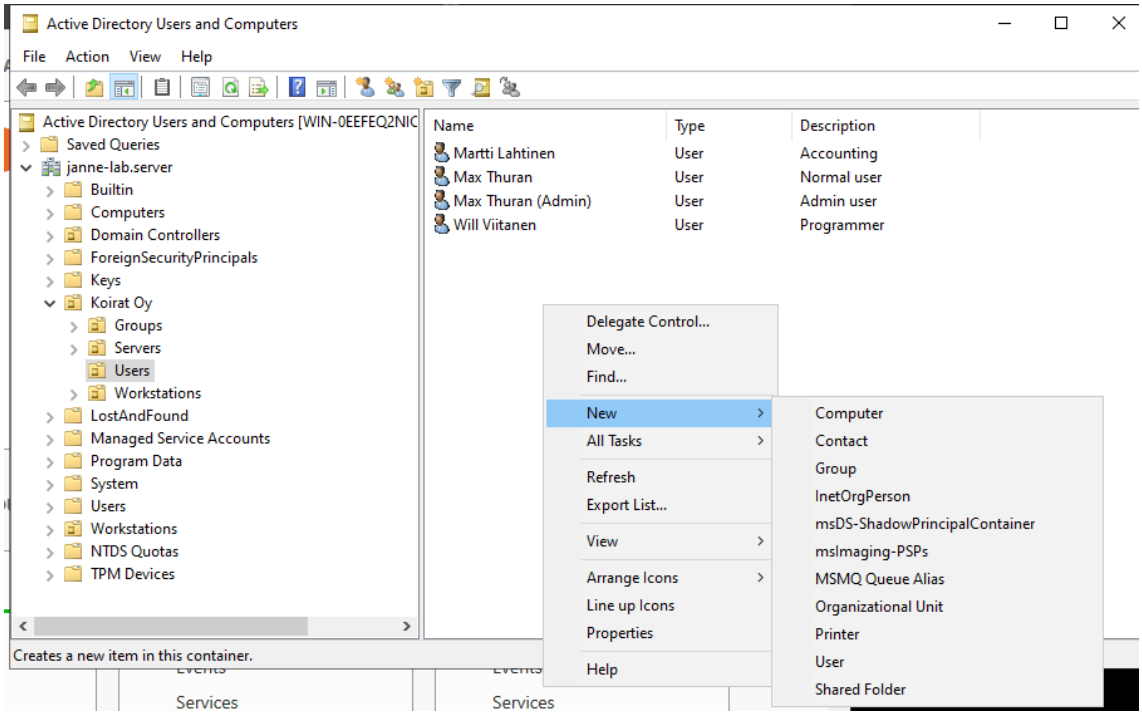
2.1 Adding groups

To add groups, go to organizational unit where you want to add them and either right click and choose New->Group or you can add the group with button on the top tool bar.



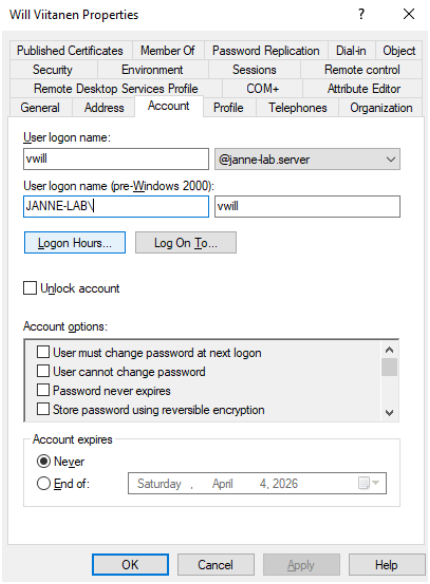
2.2 Adding users and making restrictions

New users can be added to organizational unit by right clicking and choosing New -> User.

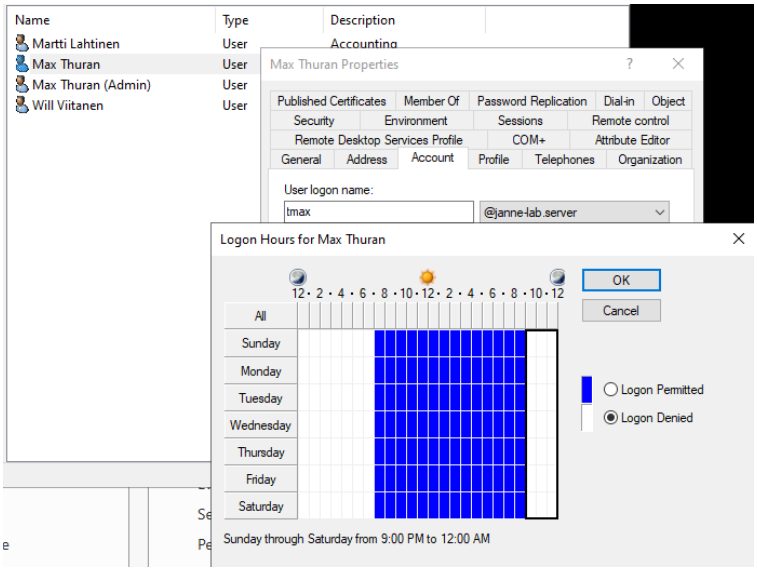


2.2.1 Logon Hours restriction

When you have added new users, you can modify them by right clicking the name and choosing properties. From the properties window you can find Account-tab and there is button to modify Logon Hours.



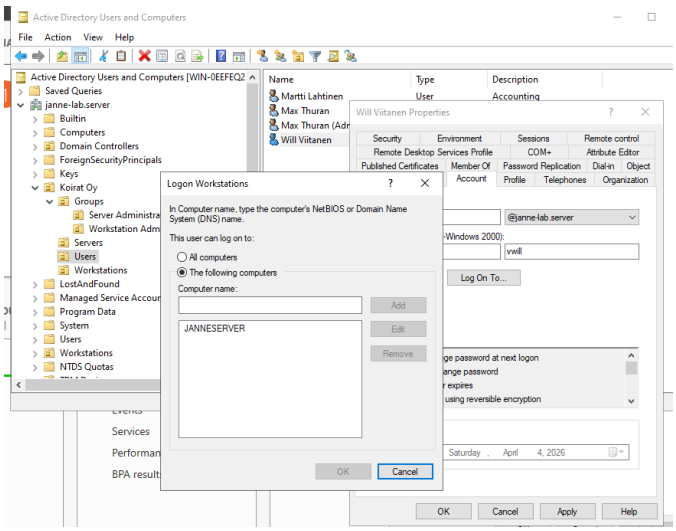
From the logon hours page, you can choose days and hour which time the user can logon.



In default the times are all available. To modify this, you can paint the time with mouse and then press from the right either “Logon Permitted” or “Logon Denied” and it will be modified and then OK.

2.2.2 Logon restrictions to specific named devices

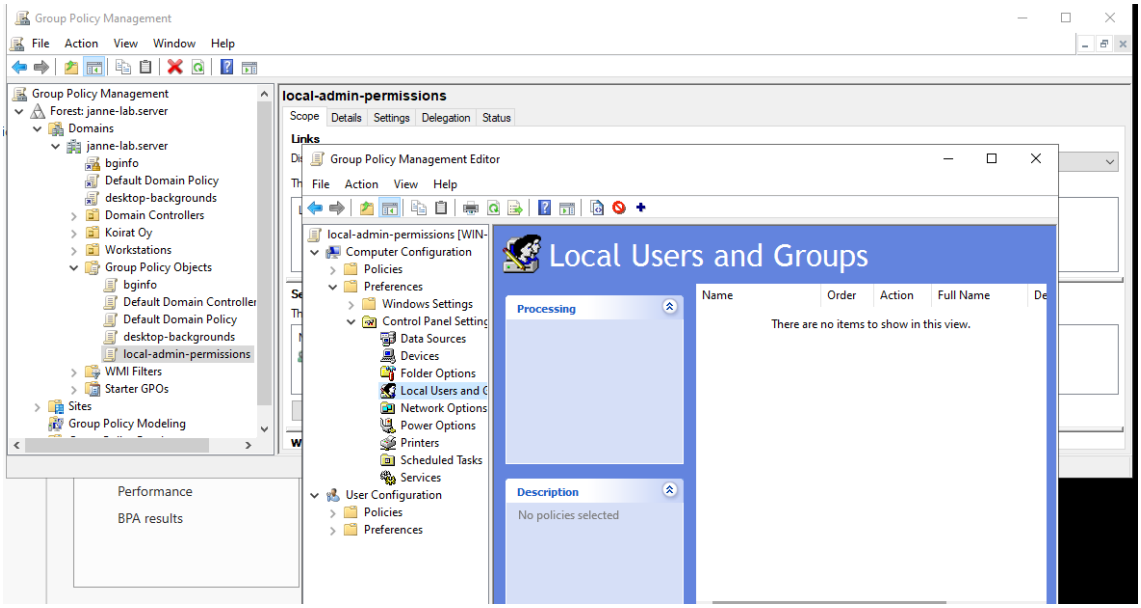
You can restrict user to logon only the specific named devices by pressing the username with right click and choosing properties. From the properties window you choose Account-tab and there is button called “Log On To…”.



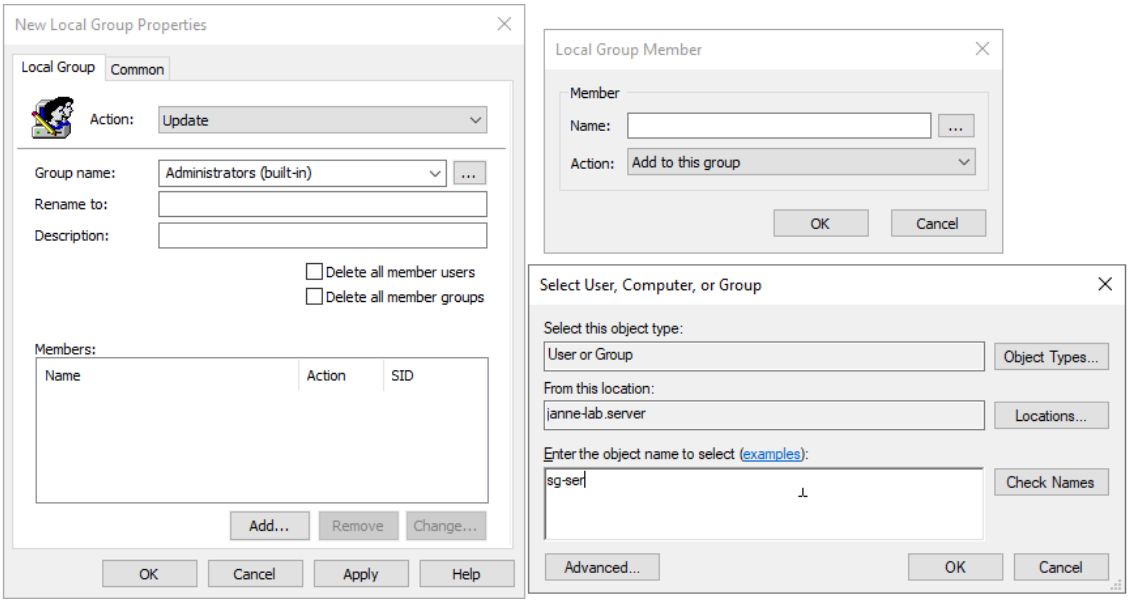
From the Logon Workstations window, choose “The following computers” and give the name of the computer you want to give the user an access. Computer name must be computer’s NetBIOS or Domain Name.

2.2.3 Adding Group Policy permissions

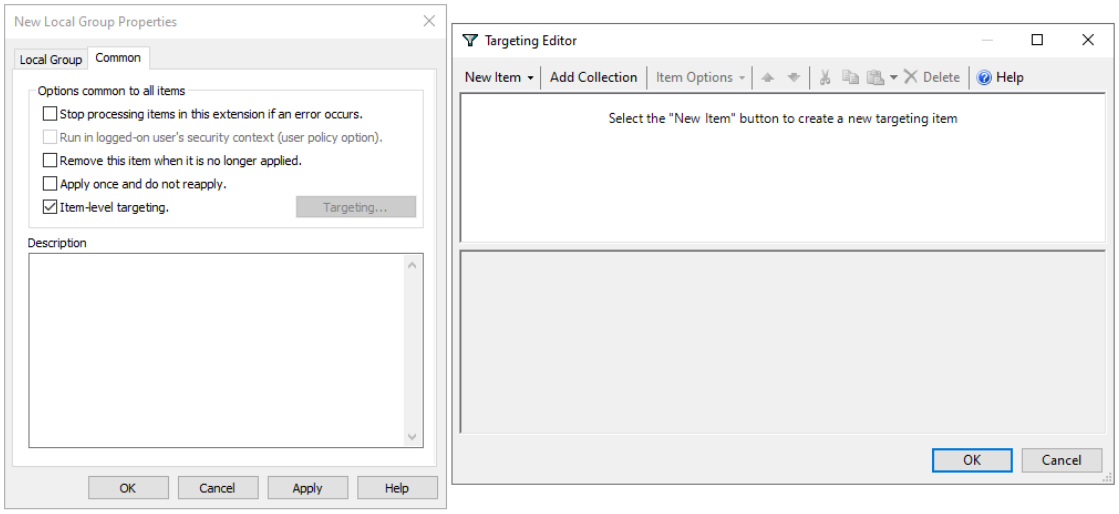
From the server manager choose tools->Group Policy Management. Here you add new group policy object by right clicking the Group Policy Objects and choose New. After it is made, you can edit it with right click. In the Group Policy Management Editor choose Computer-Configuration -> Preferences -> Control Panel Settings -> Local Users and Groups.



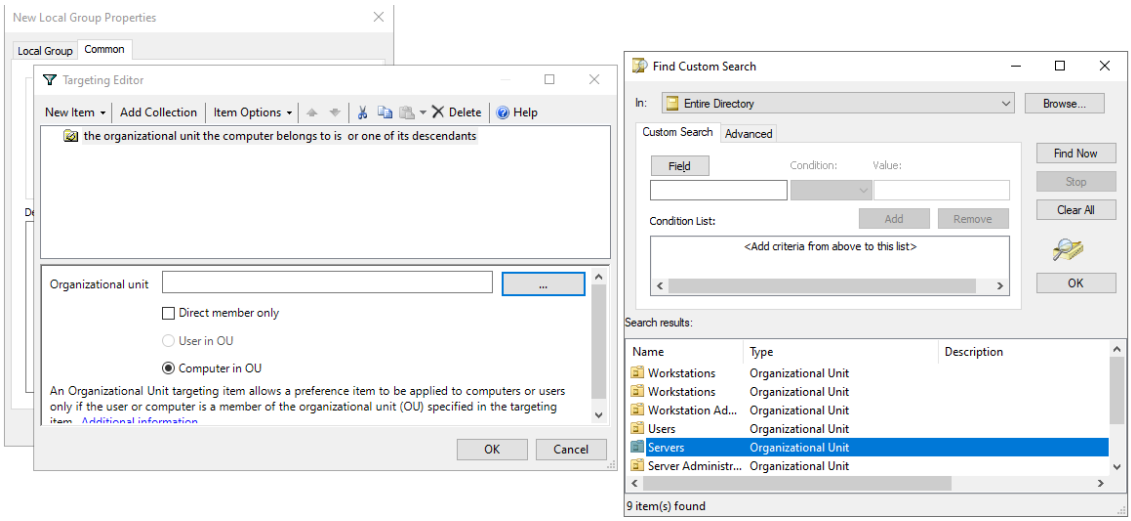
Add new Local Group by right clicking and from there select Action: Update and Group Name: Administrators (built-in). Press “Add” button and then press three dots.



Type partially the group name you have made earlier and press “Check Names”. There you can choose group you want. after the group is added, choose the “Common” tab from the top and then tick “Item-level targeting” and press button “Targeting…”.

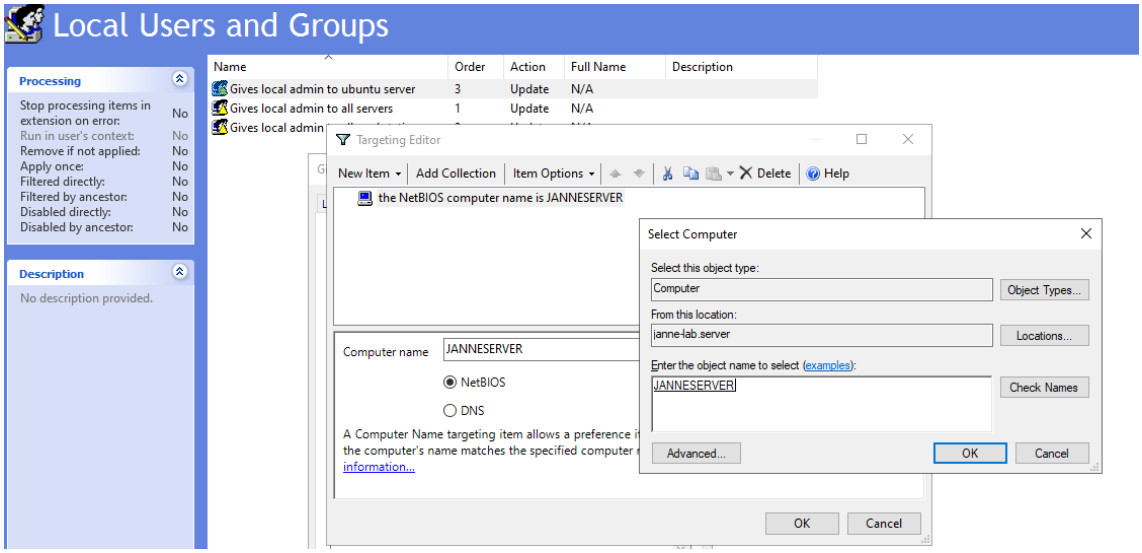


Then from the left-up corner press New item -> Organizational Unit and then press three dots button from the right to open Find Custom Search.

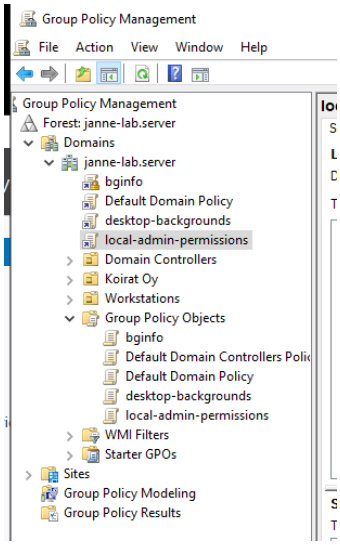


And choose the organizational unit you want use from the bottom list.

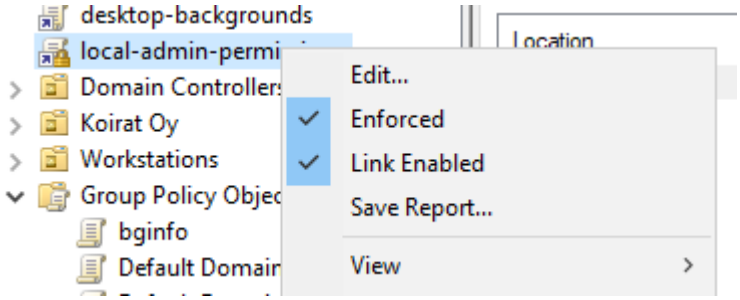
If you want to give permission only to one computer/server, press from the Targeting Editor button New Item -> Computer Name and then search the computer you want with the three dots button.



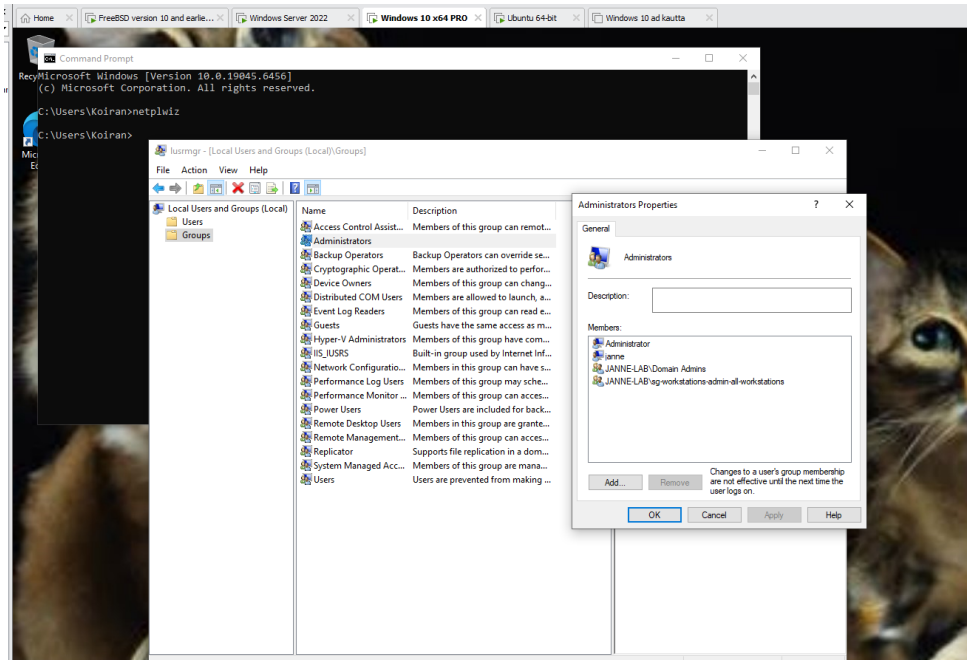
Then in the Group Policy Management drag the Group Policy Object you just made under your domain.



With right click make sure the “Enforced” is on.



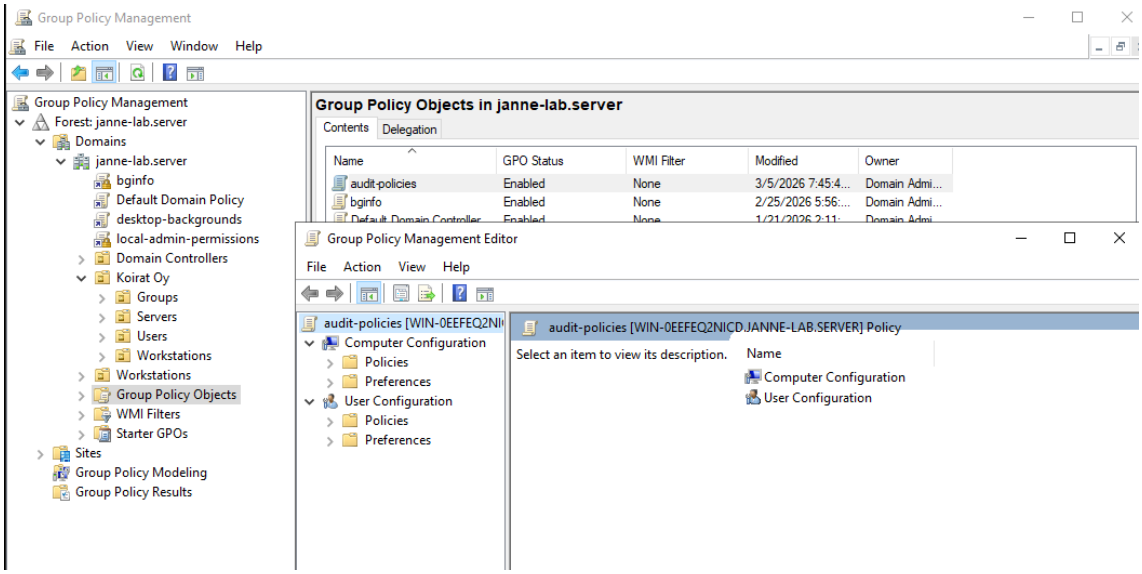
And now if you login for example to your Windows 10 workstation, and go to see from the Local Users and Groups the Groups -> administrators, there should be now new group added.



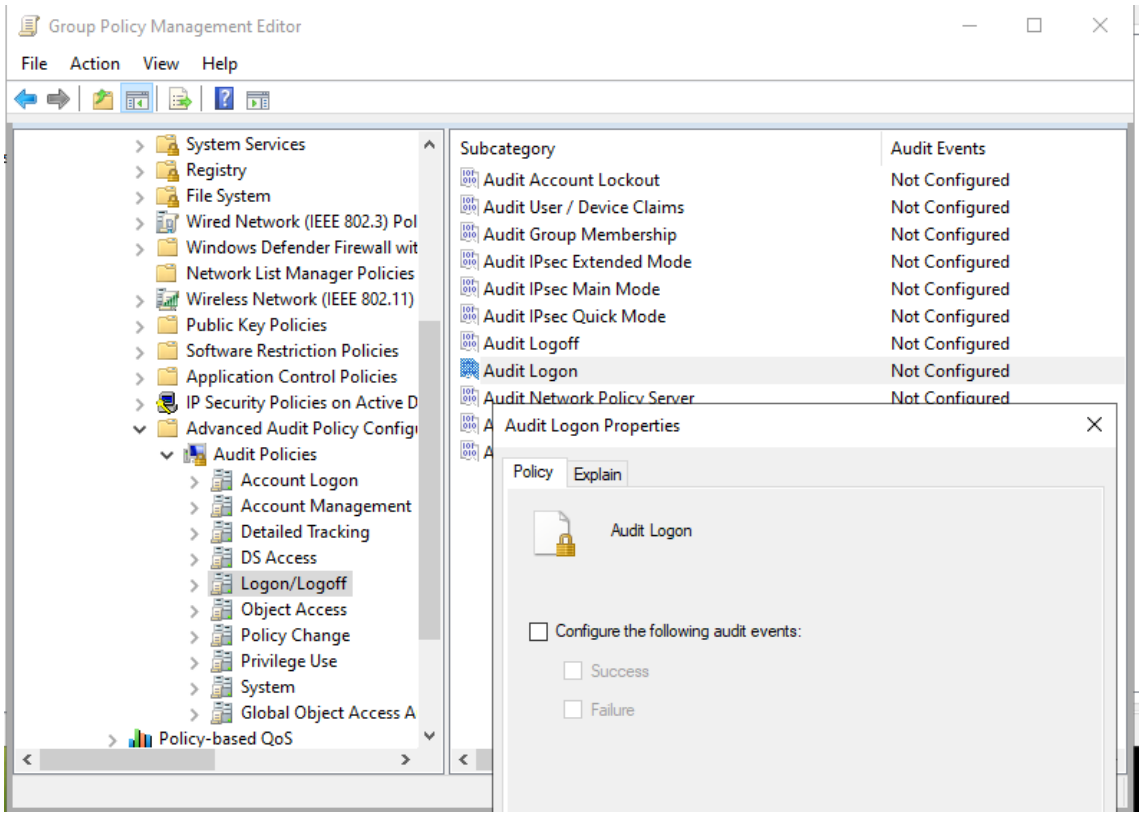
If the group is not there, give in the command prompt a command: gpupdate /force and then check again.

2.3 Logon auditing

Logon auditing in Active Directory means that you can track successful and failed logon attempts and the attempts are recorded to security event log. To add the logon auditing, from the server manager press Tools -> Group Policy Manager. Then add a new group policy object by right clicking the Group Policy Objects and choose New. Add a name, for example audit-policies. When new policy is made, right click it to edit.



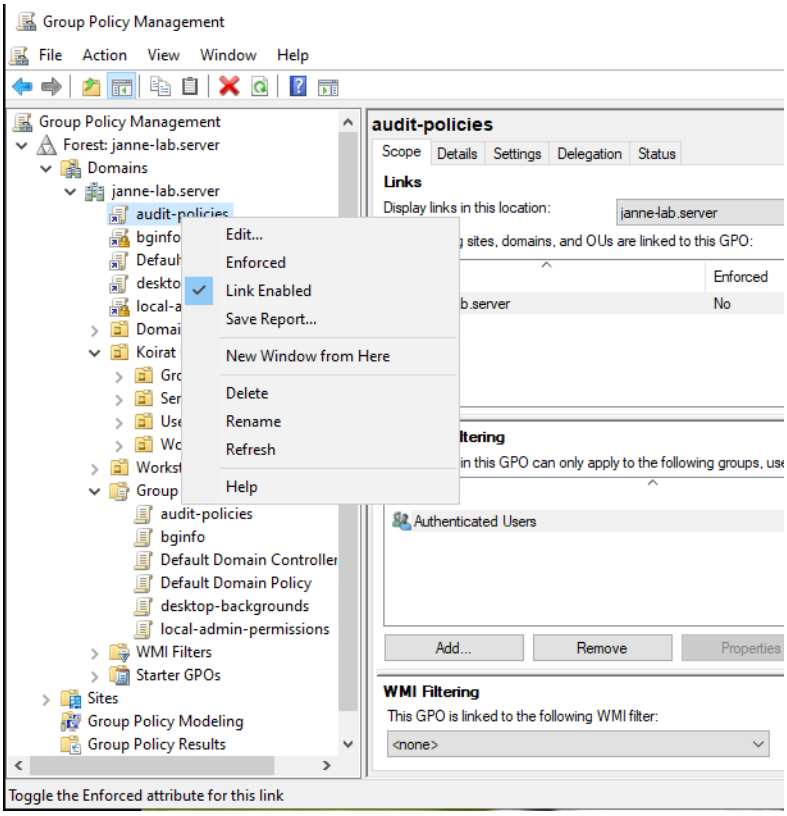
From the Group Policy Management Editor choose Computer Configuration - > Policies -> Windows Settings -> Security Settings -> Advanced Audit Policy Configuration -> Logon/Logoff -> Audit Logon.



Tick all three options: **Configure the following audit events**, **Success** and **Failure**. And do the same thing to the **Audit Account Lockout**. These two options keep a track of the local login attempts.

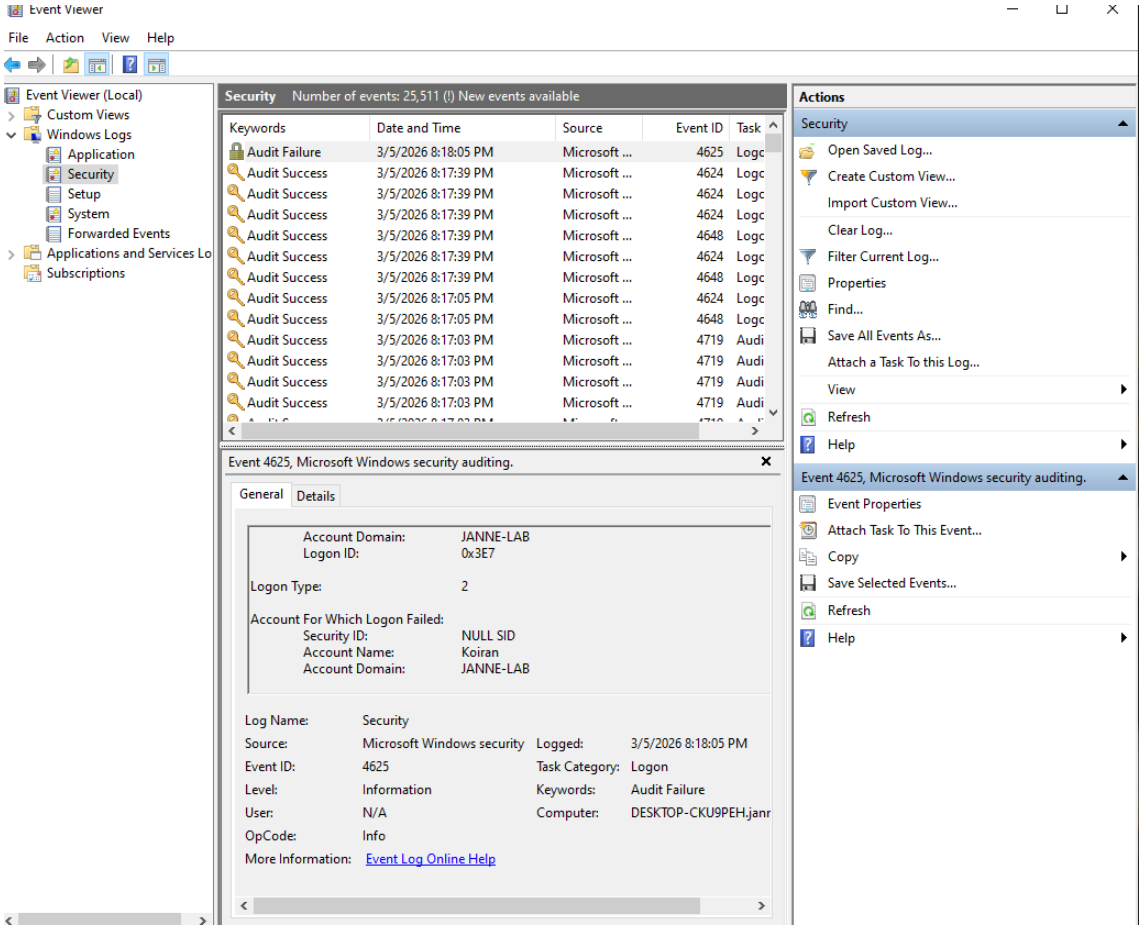
Then from the Account Logon activate: **Audit Credential Validation** which keeps track of the password verification by the domain controller, **Audit Kerberos Authentication** service which tracks Kerberos authentication activity and **Audit Kerberos Service Ticket Operations** which logs when Kerberos service tickets are requested and used to access services in an Active Directory domain.

Now the policy is ready, and you can add it to your domain by dragging it in the Group Policy Management window under your domain. And with right click enforce it on.



Now you can test this with Windows 10 workstation by running gpupdate in comman line and logoff. Then give false password to get a failed login and after

that login with real credentials. You can see the failed and successful Audits from the Event Viewer, where you need to choose Windows Logs -> Security.



Here you can see in which account was tried to login.